Industrial Ergonomics 414
ECSA GA2 Project

Group C

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1. Environmental temperature and humidity

Problem Statement: Thermal stress in hot, humid work environments can significantly impact worker performance, safety, and health leading to fatigue, heat stress, and reduced efficiency. As these conditions often can't be fully eliminated, this complex engineering problem requires a multidisciplinary approach: combining environmental data, physiological thresholds, and ergonomic principles to assess risk and develop practical, data-driven interventions that improve comfort and reduce exposure.

Methodology: Data is collected in two simulated environments: **State A** (hot, humid workspace) and **State B** (cooler rest area). Measurements are taken every five minutes using the RSPro IM-730 handheld hygrometer until readings stabilized (<0.5% change). Since the device lacks a built-in globe thermometer, a DIY black globe is created by placing the sensor inside a black cylindrical container (Figure 11) to estimate radiant heat. This study required various engineering skills:

- Mathematical: Temperature and humidity data are graphed and analyzed. WBGT is calculated using the standard equation indoors, and stability is confirmed through interval percentage changes.
- Scientific: Heat transfer principles including conduction, convection, radiation, and evaporation, aid the understanding of thermal stress in the work environment and supported the DIY globe's use in estimating radiant heat exposure.
- Fundamental Engineering: Ergonomic models from *Niebel's Work Design* are used to relate WBGT and metabolic heat to exposure limits. Based on *Grandjean's Heavy Worktables*, the worker is assumed to produce 350 kcal/hr during seated moderate arm work, with 7 hours in State A, 1 hour in State B, while acclimatized.
- Specialist Engineering: The RSPro IM-730 (±5% accuracy) measures dry bulb, wet bulb, and relative humidity. Insights from Chapter 6 of *Niebel's Work Design* informed evaluation of thermal challenges faced by Uber drivers and shaped ergonomic and environmental strategies to improve comfort and safety.

Results and Recommendations: As per the results shown in Figures 9 and 10, peak work hour measurements revealed WBGT values over 30°C, indicating a moderate to high risk of heat stress. According to Figure 6.13 in *Niebel's Work Design*, this would require a 75% rest allowance, significantly limiting productivity. The cooler rest area effectively reduced WBGT by 7–10°C, allowing for recovery. To mitigate thermal strain, several interventions are recommended. Ventilation via air conditioning or open windows can improve airflow, reduce humidity and temperature, and enhance convective and evaporative heat loss. Work-rest cycles should be adjusted using environmental data to balance productivity and safety. In mobile work contexts like Uber driving, tools such as air conditioning, reflective sun shields or tinted windows, and regular cool-off breaks can lower exposure and extend safe working time in State A. This study demonstrates how engineering judgment can address thermal stress in complex environments. Using first-principles reasoning and interdisciplinary insight, it aligns with ESCA GA2 outcomes by assessing risk and applying practical, evidence-based solutions to improve safety and efficiency.

1. Noise

Problem Statement: This study investigates how ambient conditions affect the sound pressure level (SPL) of a stationary electric generator (Honda EM 3500S), measured at a fixed distance of 1 m. According to *Niebel's Methodologies, Standards, and Work Design*, prolonged exposure to noise levels above 85 dBA can lead to permanent inner ear damage, reduced focus and fatigue. This is a complex ergonomic issue and demonstrates ECSA GA2 since SPL varies not only with distance and power of the noise source, but also with ambient noise levels and individual operator sensitivity. This makes it difficult to apply a universal solution.

Methodology: Tests are conducted in an open area to ensure accurate readings. SPL readings are measured 1 m from the noise source when it is active and then again when it is inactive. These measurements are repeated at different times (noon, 20h00 and 23h00) to study ambient effects. A calibrated sound level meter is used, fixed at 1.5 m above ground for consistency. The average SPL of the active and inactive noise source at different times are recorded and visualized in graphs (Appendix: Table 1, Figure 1).

- Mathematical: Used to indicate how different ambient conditions affect SPL with the source on and off. Visualize the results in the form of a graph.
- Scientific: Understanding the theory of sound wave propagation in free-field conditions to confirm that measured results align with theoretical predictions.
- Fundamental Engineering: Interpret data to identify safe SPL and ergonomic risk thresholds.
- Specialist Engineering: Refer to SANS 10083 and *Niebel's Work Design* to assess acceptable exposure durations and suggest mitigation strategies.

Conclusion and Recommendations: A clear trend is observed. While the source is active, the ambient conditions make no difference to the SPL perceived by the operator. Ambient noise during the day, however, masks the generators sound, while at night, lower ambient levels make the generator noise more prominent. According to *Niebel's Methodologies, Standards, and Work Design*, prolonged exposure to levels above 85 dBA poses an ergonomic risk. Readings at 1 m consistently exceed this threshold when the source is active during all the ambient conditions observed. To enhance workplace safety, it is recommended that hearing protection (eg. Earplugs) be provided during operation, especially in low ambient conditions like nightshifts and that usage time be limited in line with SANS 10083 to avoid cumulative noise exposure. Implementing these measures will reduce the risk of hearing damage and improve overall operator comfort and productivity.

2. Lighting

Problem Statement: Operator performance in reading and interpreting visual material is significantly affected by lighting ergonomics, including contrast, glare, visibility, and luminance. Inadequate or excessive lighting causes visual fatigue, slower performance, and more errors, especially when font size and surface reflectance vary as well, which increases operational costs and decreases quality. Reduced contrast between target and background, glare from reflective surfaces, and uneven illumination cause lower visibility and clarity. To address this, lighting must be evaluated and optimized using engineering methods such as lux measurements and visibility equations to ensure sufficient illuminance and the right choice of light sources. This analysis supports ECSA GA2 by defining a complex problem of visual ergonomics and lighting performance.

Methodology: This experiment evaluates how different font sizes affect reading performance under varying lighting conditions, using visibility as a factor. Two participants read 100-word text samples in five font sizes (11, 14, 17, 20, 24 pt), where each text fades from black to light grey. Tests are conducted under three lighting environments: a dark room (2.8 lux), a naturally lit room (2000 lux), and outdoor sunlight (35000 lux), recording reading time, and errors (Figure 2). Lux levels are measured using a calibrated lux meter.

- Mathematical: Luminance is calculated using $L = E \cdot R / \pi$, and visibility is quantified using the Blackwell Visibility Level (VL) equation to assess the impact of lighting and contrast on readability.
- Scientific: Light reflection, luminance, and contrast principles are applied to understand how different surface reflectance's influence text visibility.
- Fundamental Engineering: Data is interpreted to identify lighting conditions that enhance reading performance and reduce visual strain for ergonomic design.
- Specialist Engineering: SANS 10114, ISO 8995-1 and Freivalds textbook 12thEd are referenced to assess compliance with recommended lux levels and to guide lighting strategies that reduce glare and improve visibility.

This approach demonstrates GA2 by applying first principles and scientific tools to develop a structured methodology for ergonomic evaluation.

Conclusion and Recommendations:

While contrast measures brightness differences, visibility level is preferred as it accounts for visual perception under low or high-glare conditions, offering a more accurate reflection of real-world legibility. The results show a consistent relationship between surface reflectance, luminance, and VL, supporting the theoretical principles outlined in Figure 6.2 of the textbook. At 2.8 lux, all reflectance levels yielded negative VL values, with black performing worst (VL = –0.74), confirming poor visibility due to minimal luminance contrast (Figure 6). At 35782 lux, VL dropped drastically for low-reflectance surfaces (VL = –86.8 for black). Under intense lighting, low-reflectance surfaces (like black ink) can paradoxically have worse visibility because their luminance does not scale up with the brighter environment, causing the text to lose clarity against a very bright background.This trend is supported by Figures 3 and 4, which show faster reading at mid-level lux, while extreme brightness reduced performance and increased errors, as seen in Figure 5.

Optimal reading conditions are found around 2000 lux, where luminance and contrast are better balanced across reflectance types (Figure 7). To maintain visual clarity, surface reflectance should follow textbook recommendations—white backgrounds (85–90%) and dark text (3–6%) as noted in Table 6.1. Lighting above 30000 lux should be managed with shading or matte finishes to control glare. Diffuse, uniform illumination is essential to avoid directional reflections and visual fatigue. Based on VL trends and task performance data, maintaining workplace lighting between 500–1000 lux offers ideal ergonomic conditions, reduces reading errors, and supports sustained visual comfort.

3. Manual tasks IoT measurement

Problem Statement: A manual task can be broken down into elements, each requiring a certain hand movement. Hand movements contribute significantly to operator fatigue and cycle time. *Niebel's Methodologies, Standards, and Work Design* classify these hand movements as effective or ineffective using Therbligs. The aim is to minimize the ineffective Therbligs or if possible, eliminate them. However, this is a complex problem because it is challenging to identify and classify these movements, especially when operators work at high speed. Human observation is also subjective and therefore a data-driven engineering approach is required to quantify and model the motion. In this way, hand movements can be objectively identified and classified, and the manual tasks can be redesigned and optimized.

Methodology: Data is collected in a controlled workspace environment using a wearable IoT device, attached to the operator's forearm. The wearable is equipped with a 6-axis accelerometer and captures rotational and linear movements during three separate movements:

a. The operator's hand rotation (yaw, pitch and roll) around the x, y and z axes is tested to find the maximum rotation range in each direction.
b. Moving an object linearly to quantify acceleration and displacement.

c. The repetitive task of lifting a glass is tested to identify the pattern in acceleration that defines that movement. From this, the signature is extracted.

The wearable measures movement every 20 milliseconds but only returns the maximum values over the movement period and from this, various graphs can be configured and conclusions drawn.

The problem requires various skills:

- Mathematical: Mathematics is used to calculate motion statistics from the raw accelerometer data.
- Scientific: Basic scientific understanding of human anatomy and the physics of motion are used interpret the data produced by the sensor.
- Fundamental Engineering: Basic experimental design is used to detect movement patterns and critically analyse the data produced.
- Specialist Engineering: Table in *Niebel's Methodologies, Standards, and Work Design* is applied to classify each Therblig. The task is redesigned for efficiency and safety.

Conclusions and recommendations:

a. Figure 12 is plotted for one measurement of task A and maximum rotation for each movement along the x, y or z-axis is identified. For example, for the "yaw" movement, the maximum angular velocity is 39.88 degrees/second and this rotation occurs in the z-direction.
b. For task B, figure 13 shows the maximum recorded accelerations along the orthogonal axes for three separate measurements. All three measurements show maximum acceleration (m/s^2) in the z-direction.
c. Figures 14 and 15 are drawn to display maximum accelerations and angular velocities along the orthogonal axes for three measurements respectively. The signature of the task is identified by the axis along which maximum measurements occurred. The three measurements indicate that maximum acceleration occurs along the x-axis and maximum rotation occurs on the z-axis. This is therefore the signature for the task of lifting a glass.

IoT-based task measurement should be implemented in workspaces where highly repetitive tasks are performed. In this way, data can be collected and analysed and improvements can be made in the workspace. A trained operator can perform a particular task and the signature for that task can be extracted. Then, tasks can be redesigned if necessary to eliminate Therbligs or new operators can be tested to observe if the task is being performed correctly. Inefficient or unsafe techniques can be easily identified. ECSA GA2 is demonstrated in this analysis by using first principles to identify Therblig's and quantifying the ergonomic factor by means of data capture.

Appendix A:

Table 1: Sound Pressure Level

Sound Pressure Level [SPL(dBA)]		
Ambient Condition	**Electric Generator State**	**Reading [dBA]**
Noon	Active	92
	Inactive	38.7
20h00	Active	92
	Inactive	38
23h00	Active	92
	Inactive	37.9

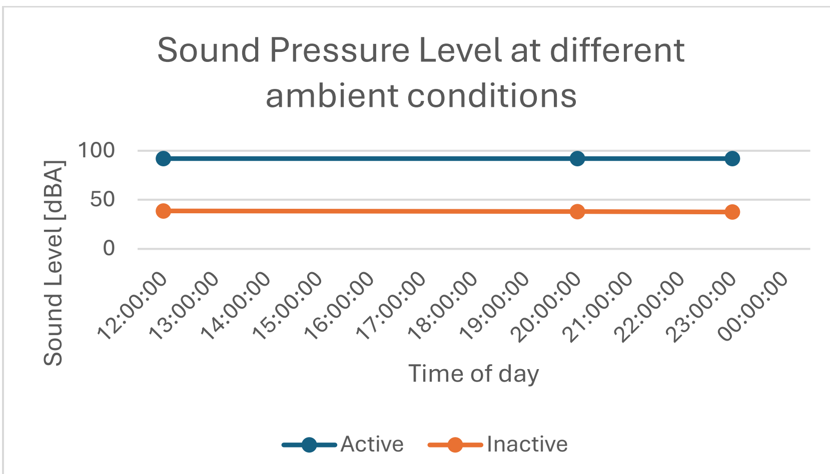

Figure 1: Sound Pressure Level at different ambient conditions

		TIME (seconds)		TIME (seconds)	
LUX	FONT	PERSON 1	ERRORS P:	PERSON 2	ERRORS P2
2087	11	44.01	1	43.31	
2087	14	42.5		41.29	
2087	17	41.16		40.01	
2087	20	39.4	1	38.2	
2087	24	37.51		37.01	2

		TIME (seconds)		TIME (seconds)	
LUX	FONT	PERSON 1	ERRORS P1	PERSON 2	ERRORS P2
2.8	11	45.45	4	45.05	3
2.8	14	43.5		42.36	2
2.8	17	43.1	1	42.06	
2.8	20	40.58		41.47	
2.8	24	39.55		40.14	1

LUX	FONT	TIME (seconds) PERSON 1	ERRORS P1	TIME (seconds) PERSON 2	ERRORS P2
35782	11	46.12	3	46.49	4
35782	14	45.1	2	44	
35782	17	42.5		41.9	3
35782	20	41.3	1	39.46	
35782	24	39.48	1	38.29	1

Figure 2: Results of reading times (sec) and errors made at 3different lux readings

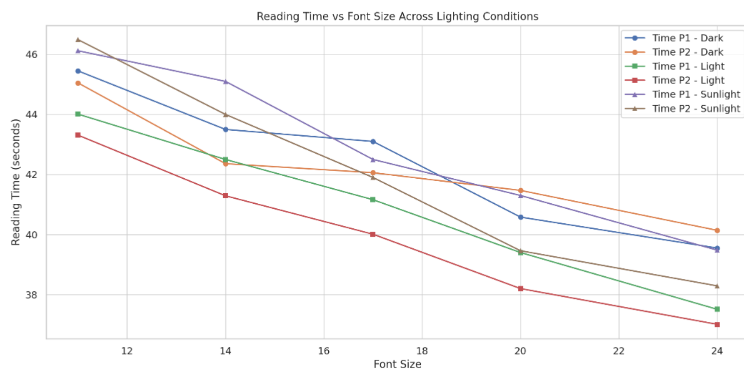

Figure 3: Comparison of reading times across font sizes under dark, natural, and sunlight conditions for both participants

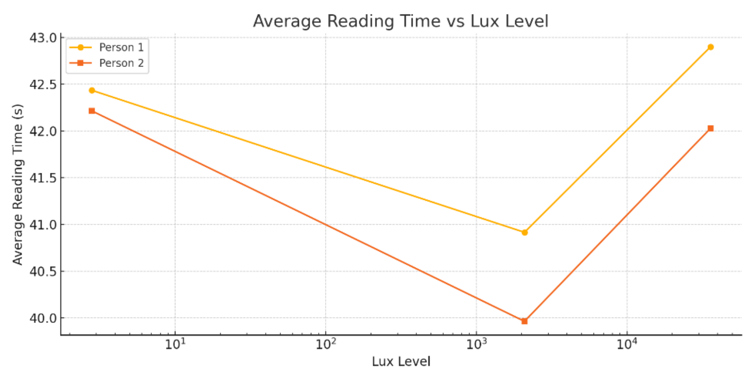

Figure 4: Average reading time at each lux level.

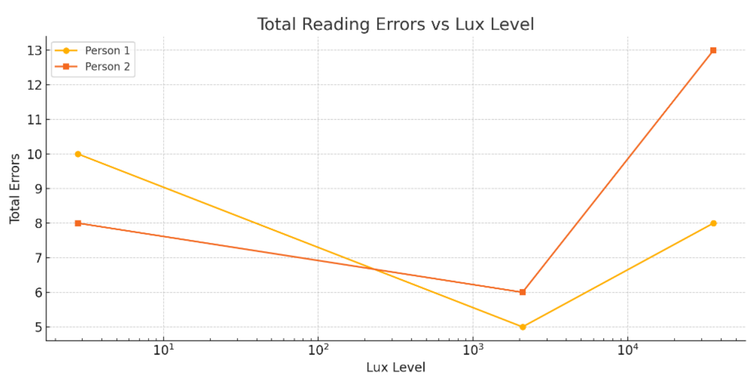

Figure 5: Total reading errors for each lux level.

Reflectance	VL @ 2.8 lux	VL @ 2087 lux	VL @ 35782 lux	LT @ 2.8 lux	LT @ 2087 lux	LT @ 35782 lux	LB (2.8 lux)	LB (2087 lux)	LB (35782 lux)
0.75 (light grey)	-0.09918	-2.79585	-11.57567	0.668449	498.2334	8542.303	0.757576	564.6645	9681.277
0.55 (med grey)	-0.29753	-8.38646	-34.72702	0.490196	365.3711	6264.356	0.757576	564.6645	9681.277
0.3 (dark grey)	-0.54548	-15.37516	-63.66621	0.267387	199.293	3416.921	0.757576	564.6645	9681.277
0.1 (black)	-0.74384	-20.96613	-86.81755	0.089127	66.43112	1138.974	0.757576	564.6645	9681.277

Figure 6: Visibility and Luminance Table

Contrast	0.75	0.117647	0.117647	0.117647	light grey
	0.55	0.352941	0.352941	0.352941	med grey
	0.3	0.647059	0.647059	0.647059	dark grey
	0.1	0.882353	0.882353	0.882353	black

$$\text{Contrast} = (L_{\max} - L_{\min})/L_{\max}$$

Figure 7: Contrast Values at different Reflectance

Lighting in the workplace significantly affects employee performance, mood, and overall well-being. Natural light is often preferred, as it reduces eye strain and increases productivity. However, artificial lighting must be well-designed to prevent glare and shadows. Cool white light can promote alertness, while warm lighting is more relaxing. Poor lighting contributes to fatigue, headaches, and even mistakes. Adjustable lighting, including task lamps, allows individuals to customize their environment. Light placement and intensity also influence comfort, especially when using screens. Balanced lighting enhances focus and accuracy. Overall, effective illumination supports a healthier, more efficient working environment for all types of tasks.

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Accessibility: Investigate

Figure 8: 100 word reading script using different tones and sizes

Date	Time	Location	State (A/B)	Tdb (Dry Bulb, °C)	Tnwb (Wet Bulb, °C)	RH (%)	Globe Temp (Tg, °C)	WBGT (Indoors)	% Change in Tdb	% Change in Tnwb	% Change in RH
18/04/2025	16:05	Worksite	A	32.2	28.1	38.8	33.1		0%	0%	0%
18/04/2025	16:10	Worksite	A	33.0	29.0	43.9	33.9		0.80%	0.90%	5.10%
18/04/2025	16:15	Worksite	A	33.1	29.9	51.0	34		0.10%	0.90%	7.10%
18/04/2025	16:20	Worksite	A	33.1	29.8	51.3	34	31.06	0.00%	0.10%	0.30%
18/04/2025	17:25	Rest Area	B	25.6	21.6	42.8	25.8		0%	0%	0%
18/04/2025	17:30	Rest Area	B	25.5	21.5	43.0	25.8		0.10%	0.10%	0.20%
18/04/2025	17:35	Rest Area	B	25.5	21.5	43.1	25.7		0.00%	0.00%	0.10%
18/04/2025	17:40	Rest Area	B	25.4	21.4	43.1	25.6	22.66	0.10%	0.10%	0.00%

Figure 9: Tabulation of Temperature and Humidity results

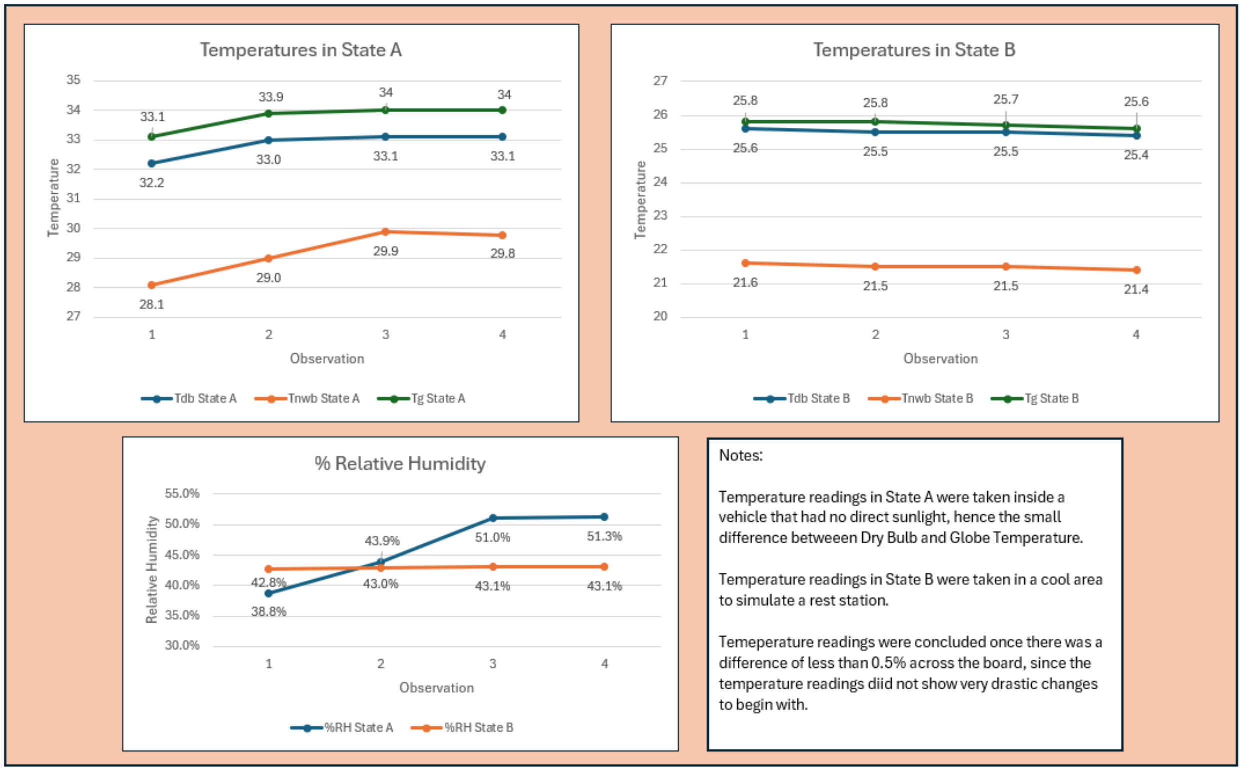

Figure 10: Graphs of the change in Temperature and Humidity readings

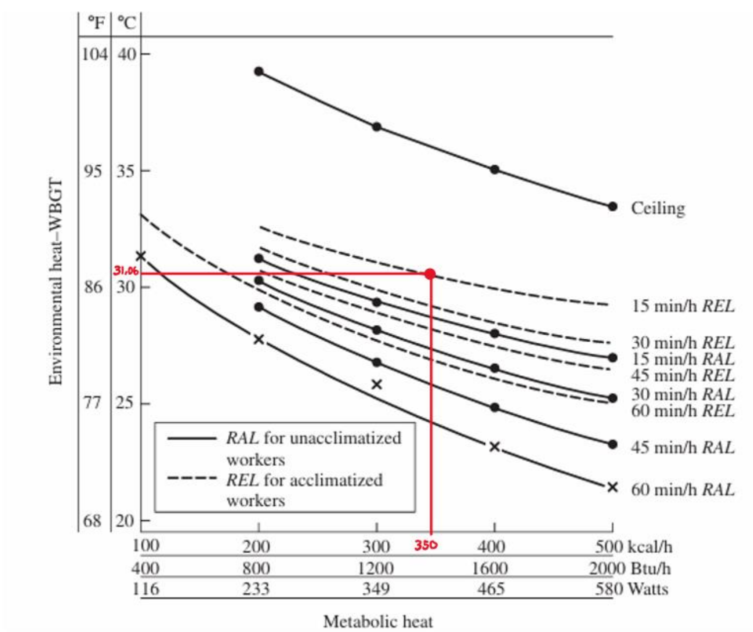

Figure 6.13 Recommended heat stress levels based on metabolic heat (1-h time-weighted average), acclimatization, and workrest cycle.

Figure 11: DIY Globe temperature cylinde

Maximum recorded angular velocity (degrees/second) along X, Y and Z axes for yaw, pitch and roll movements

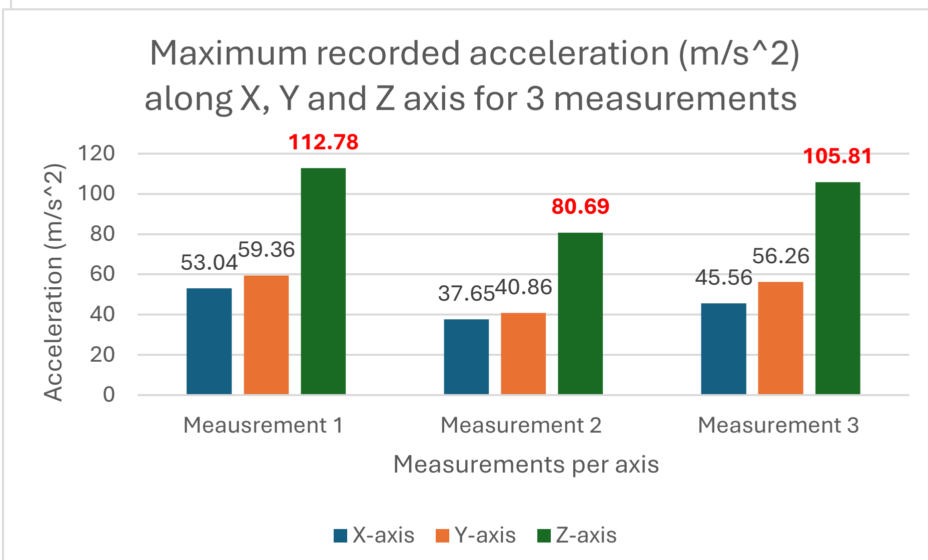

Figure13: Results for task b

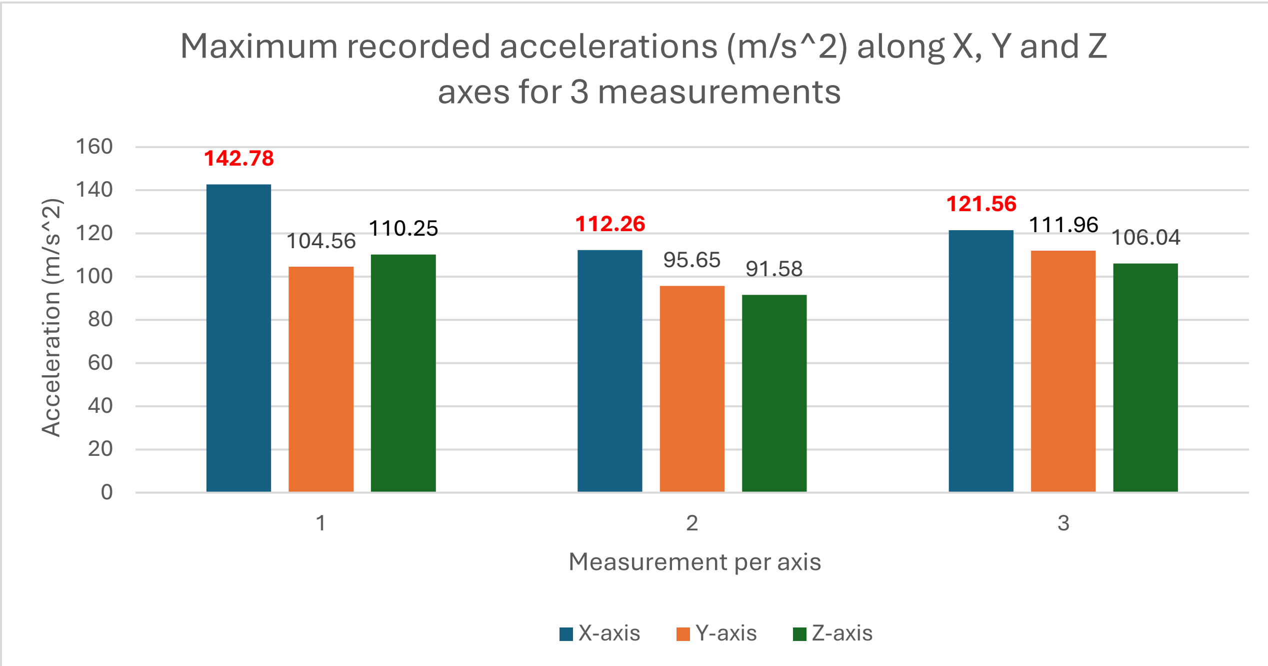

Figure 14: acceleration results for task c

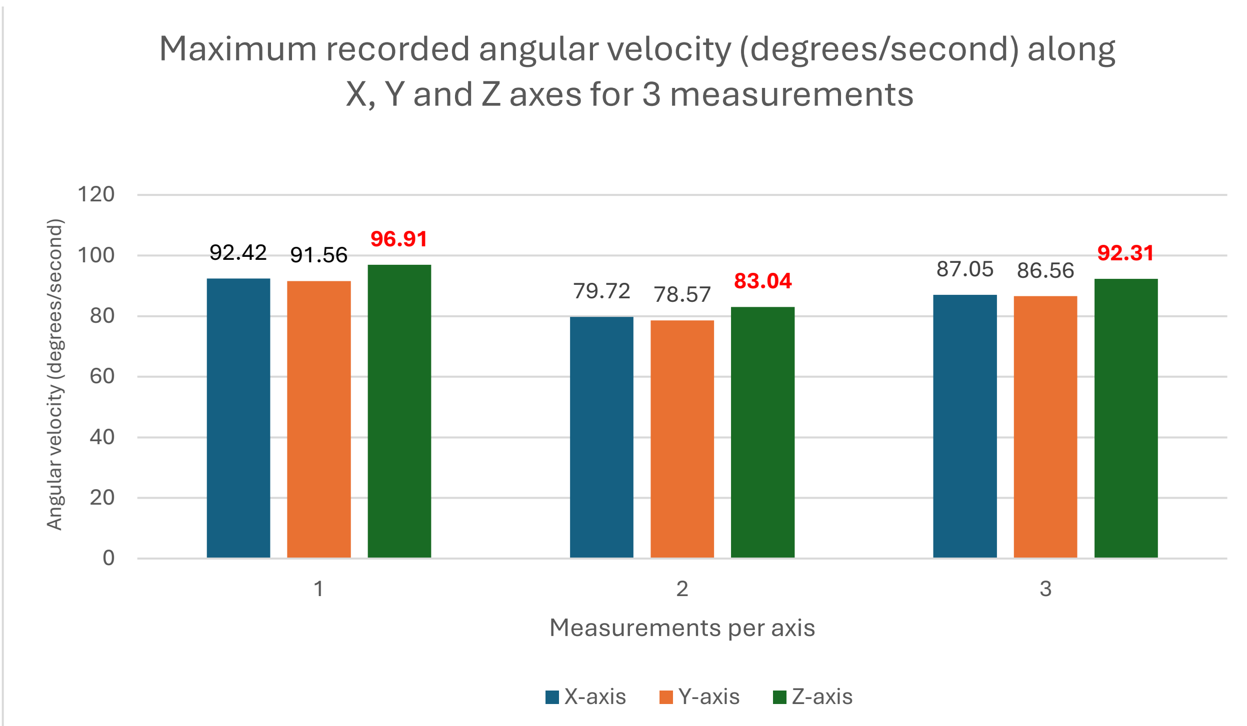

Figure 15: angular velocity results for task c